

INTERNSPARK VLSI DESIGN INTERNSHIP

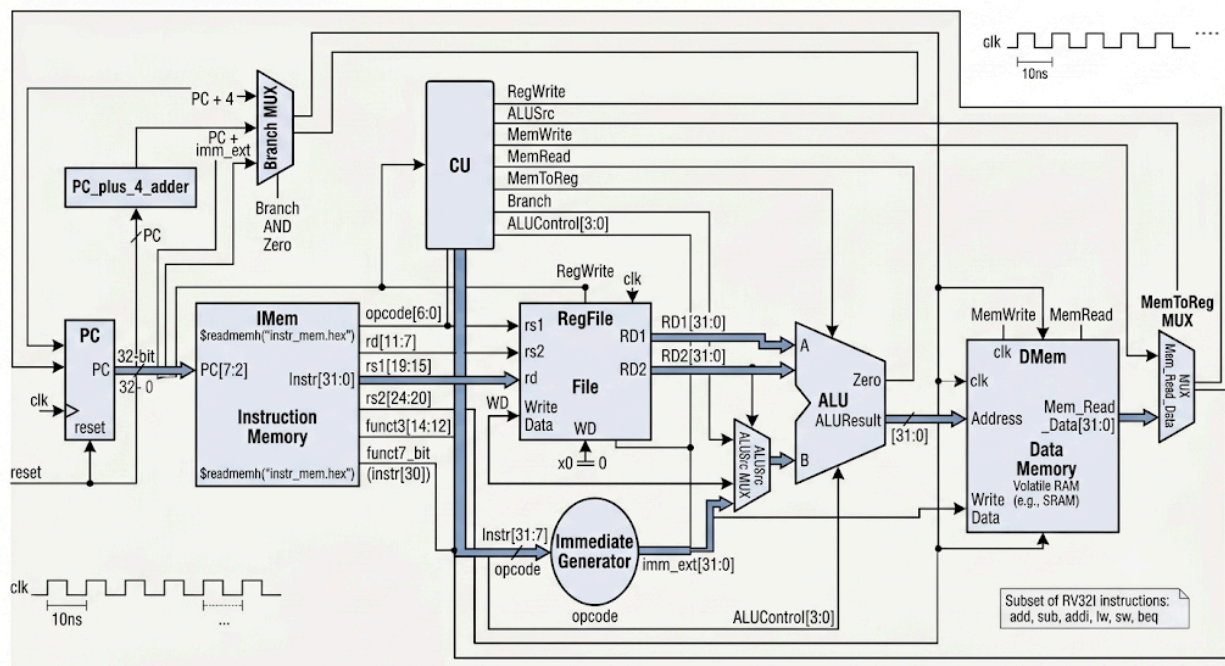
TASK 1 REPORT: SINGLE-CYCLE RISC-V DATAPATH & RTL DESIGN

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2. Introduction & Block Diagram

- **Objective:** Design, implement, and verify a 32-bit Single-Cycle RISC-V processor core using Verilog HDL. The execution flow must process an entire instruction—from instruction fetch to write-back—within a single clock cycle.
- **Architecture Overview:** The design separates the processor into a **Datapath** (the hardware elements that operate on data: PC, Memories, Register File, ALU) and a **Control Unit** (the logic that orchestrates which elements are active based on the instruction opcode).



3. Supported Instruction Set

Instruction Type	Mnemonic	Opcode (Hex/Binary)	Description
R-Type	ADD	0110011	Adds two registers: $\$R[rd] = R[rs1] + R[rs2]$
R-Type	SUB	0110011	Subtracts two registers: $\$R[rd] = R[rs1] - R[rs2]$

I-Type	ADDI	0010011	Adds immediate value to register: $\$R[rd] = R[rs1] + \text{sign_ext}(\text{imm})\$$
I-Type	LW	0000011	Loads a word from Data Memory to register
S-Type	SW	0100011	Stores a word from register to Data Memory

4. Hardware Module Descriptions

- **Program Counter (PC)**

A 32-bit register that holds the memory address of the current instruction. On every rising edge of the clock, it increments by 4 ($\$PC = PC + 4\$$) to point to the next sequential instruction.

- **Instruction Memory**

A read-only memory structure initialized using your external hex file (`instr_mem.hex`). It takes the 32-bit PC as an address input and outputs the 32-bit machine code instruction.

- **Register File**

A dual-read, single-write storage bank containing 32 general-purpose registers (`x0` to `x31`).

- Register `x0` is hardwired to `32'h00000000`.

- It reads two source registers (`rs1`, `rs2`) asynchronously and writes back data to the destination register (`rd`) synchronously on the positive clock edge when `RegWrite` control signal is high.

- **Immediate Generator (Imm-Gen)**
Extracts and reformats scrambled immediate bit fields out of I-type or S-type instructions, sign-extending them into a uniform 32-bit value.
- **Arithmetic Logic Unit (ALU)**
The core combinational block that executes calculations. It reads operational selectors from the ALU Control unit to perform mathematical operations (**ADD**, **SUB**, **AND**, **OR**) on its inputs.

Main Control Unit

The brain of the processor. It evaluates the 7-bit opcode field of the instruction to drive control signals across the datapath:

- **ALUSrc**: Selects between Register File data or Immediate value for the second ALU input.
- **RegWrite**: Enables writing data back to the destination register.
- **MemtoReg**: Determines if the data written to the Register File comes from the ALU or Data Memory.

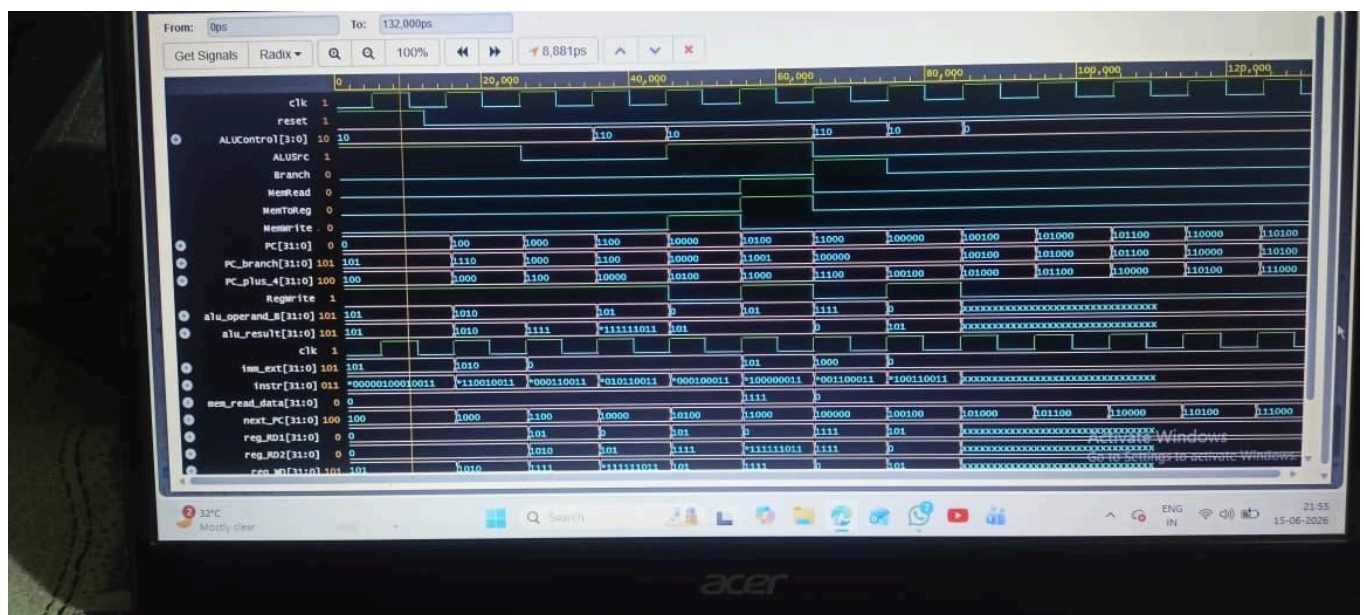
5. Test Program Details (Hex & Assembly Translation)

PC (Binary)	PC (Dec)	Machine Instruction (Binary)	Decoded Assembly	Technical Operation & Register Updates
0	0	000000100010011	addi x1, x0, 5	Load immediate value 5 (101 b) into register x1. alu_result and reg_WD confirm 101.

100	4	110010011	addi x2, x0, 10	Load immediate value 10 (1010 b) into register x2. reg_WD updates to 1010.
1000	8	000110011	add x3, x1, x2	Adds x1 (5) and x2 (10). alu_result updates to 1111 (15 decimal).
1100	12	010110011	sub x4, x1, x2	Subtracts x2 from x1 (\$5 - 10 = -5\$). alu_result displays the 2's complement negative value.
10000	16	0100011	sw x3, 0(x1)	Store Word: MemWrite goes High. The value of x3 (15) is written to data memory at the address calculated by the ALU.

10100	20	100000011	lw x5, 0(x1)	Load Word: MemRead and MemToReg both go High. Data is pulled from memory (mem_read_data = 1111) and sent to reg_WD.
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6. Simulation Results & Waveform Verification



A. Instruction Fetch & Decoding Verification

As seen in the waveform, the Program Counter (**PC[31:0]**) increments perfectly on every rising edge of the clock signal (**clk**) along sequential boundaries: 0
 $\rightarrow$ 100 (4) $\rightarrow$ 1000 (8) $\rightarrow$ 1100 (12)
 $\rightarrow$ 10000 (16) $\rightarrow$ 10100 (20). The **instr[31:0]** bus updates instantly, validating stable instruction fetch logic from memory.

B. Execution Unit (ALU & Control Logic) Verification

- **Arithmetic Verification:** During the execution of the **add** instruction (at **PC = 1000**), **ALUControl** updates to select addition, taking **101** and **1010** as inputs, outputting the correct sum **1111** (15).
- **Data Memory Handshake:** At **PC = 10000**, the **MemWrite** signal transitions to a logic high state, executing a memory write. Immediately following at **PC = 10100**, **MemRead** and **MemToReg** switch high, confirming that data memory contents are multiplexed onto the register write data bus (**reg_WD**).